- Hyper-rectangular regions, cf. figure 9.26; here the geometric interpretation is to query points whose values are in certain ranges.

When we deal with spherical regions, we are in a similar situation as the nearest neighbor search: we need to include all the points within a certain distance from a given point. It's like performing a NN-search, where we never update the distance to the nearest neighbor, and instead of keeping track of only one point (the *NN*), we gather all points closer than that distance.

In figure 9.25 you can see how we are going to prune branches: when we are at a split, we will certainly have to traverse the branch on the same side of the center of the search region P; for the other branch, we check the distance between P and its projection on the split line: if that distance is lower than or equal to the search region's radius, it means that there is still an area of intersection between that branch and the search region, and so we need to traverse the branch; otherwise, we can prune it.

